

ASSIGNMENT NO .1.

Title: Identification and Selection of Problems

Objective: To identify the real-world problems for design thinking

The Four Ws

- **Who :-** Daily urban commuters — office workers, students, and travelers who rely on multiple modes of transport.
- **What :-** Urban travel systems operate in disconnected segments (bus, metro, train, auto, etc.) without unified coordination, real-time integration, or predictive updates.
- **Where :-** In metropolitan cities during daily commuting across multiple transport systems.
- **Why :-** This problem is important because the lack of coordination makes commuting unpredictable and stressful. Even small delays can disrupt the entire journey and waste valuable time.

The Five Whys

1. **Why do commuters find travel unpredictable?**
Because delays in one segment affect the entire journey.
2. **Why do delays affect the entire journey?**
Because transport systems are not synchronized.
3. **Why are transport systems not synchronized?**
They operate independently without integrated data sharing.
4. **Why is there no integrated data sharing?**
There is no centralized coordination or unified platform.
5. **Why is there no unified platform?**
Urban mobility planning focuses on individual transport modes rather than end-to-end journey experience.

Identifying the root cause —

Lack of integrated, real-time, end-to-end coordination and predictive intelligence in urban transportation systems.

Problem statement:

Urban commuters need an integrated, real-time journey coordination system that provides end-to-end visibility, predictive updates, and synchronized transport information to reduce uncertainty, stress, and delays during daily travel.

Statement:

Urban commuters face significant challenges because city transportation systems function as isolated segments rather than as a unified journey. Passengers must independently coordinate multiple modes of transport such as buses, metros, local trains, and ride-sharing services. Due to fragmented information and lack of real-time integration, even small delays in one segment can disrupt the entire trip. This absence of end-to-end visibility and predictive updates makes daily commuting stressful, uncertain, and mentally exhausting.

Discussion:

Urban transportation systems work separately instead of as one connected system. Commuters have to manage different modes of transport like buses, metros, and trains on their own. Since there is no unified platform providing real-time updates and coordination, even small delays can disturb the entire journey.

This lack of integration makes daily travel stressful, unpredictable, and time-consuming. The main issue is the absence of an end-to-end system that connects all transport services and provides accurate, synchronized information to commuters.

Submitted By:

Name :	Tanisha Borgave, Atharv Kulkarni, Ronakkumar Hurkat
Branch:	Computer Science & Engineering (Un-Aided)
Batch:	S5
PRN No:	245109072, 245109074, 245109078

Assignment 2

Title: Customer Survey and Analysis

Objective: To gather and analyze customer feedback through structured surveys to better understand their issues and needs.

Theory:

Surveys consist of a series of questions directed at individuals to collect meaningful data. They are designed to explore the thoughts, feelings, and preferences of a specific group. When targeting a large population is impractical due to time or cost constraints, a representative sample is selected to reflect the broader group's characteristics or to test specific hypotheses about their behaviors or relationships.

Advantages of Surveys:

1. **High Representativeness:** A properly selected sample can provide valuable insights into the behavior and preferences of a larger commuter population.
2. **Cost-Effectiveness:** Online surveys require minimal cost and can reach a large number of respondents easily.
3. **Ease of Data Collection:** Digital platforms allow quick distribution and automatic recording of responses.
4. **High Objectivity:** Minimal direct interaction reduces researcher bias.
5. **Standardization:** All respondents answer the same set of questions, ensuring consistency in data collection.

Disadvantages of Surveys:

1. **Inflexible Design:** Once conducted, survey questions cannot be modified.
2. **Risk of Inauthentic Responses:** Respondents may provide inaccurate or socially desirable answers.
3. **Lack of Depth:** Surveys may not capture detailed emotional or personal experiences.

Modern Survey Methods and Analysis Techniques:

Online Surveys: Conducted using platforms like Google Forms for quick and automated data collection.

Mobile Surveys: Allow participants to respond through smartphones for better accessibility.

Interactive Voice Response (IVR): Automated phone-based surveys using keypad or voice responses.

Analysis Techniques:

Quantitative Analysis: Statistical evaluation of percentage-based responses to identify patterns.

Qualitative Analysis: Thematic analysis of open-ended responses to understand commuter sentiments.

Mixed Methods: Combination of both numerical and descriptive data analysis.

Your Feedback and Its Analysis**Survey Details:**

- Sample Size: 50 Urban Commuters
- Target Group: Students and Working Professionals
- Method Used: Online Survey

Survey Findings:

- 78% use multiple transport modes daily.
- 65% frequently face delays due to poor coordination.
- 72% experience stress during commuting.
- 90% prefer a single integrated travel platform.

Google Forms Link :

https://docs.google.com/forms/d/e/1FAIpQLSeUoyuZj4jrQJnWBurGs8_hFRbvSKW0Kff3xsYfEl0pZNkIVg/viewform?usp=publish-editor

Data From Google Forms :

Respondent_ID	Name	Age	Address	Q1_Ease_of	Q2_Real_Time_Accuri	Q3_Stress	Q4_App_j	Q5_Need	Q6_Delay	Q7_Crow	Q8_Traffi	Q9_Intere	Q10_Overall	Ans
1	Aarav Patil	43	House No 14, Shivaji Nagar, Sangli	4	2	4	2	5	1	2	5	5	2	
2	Rohit Kulkarni	50	House No 5, Civil Hospital Road, Sangli	2	2	2	1	4	2	5	5	1	2	
3	Neha Deshmukh	28	House No 104, Ramnagar, Sangli	3	4	1	3	2	2	4	5	1	1	
4	Siddharth Jadhav	19	House No 16, Sanjay Nagar, Sangli	2	1	4	2	4	1	1	5	2	2	
5	Priya Shinde	60	House No 19, Sangli-Miraj Road, Sangli	2	5	5	1	3	2	4	1	5	2	
6	Omkar Joshi	35	House No 52, Padmavati Nagar, Sangli	3	5	3	2	3	2	1	4	4	5	
7	Sneha Pawar	60	House No 64, Madhavnagar Road, Sangli	3	4	5	5	1	4	4	2	3	2	
8	Aditya Chavan	46	House No 193, Khanbhag, Sangli	3	2	5	2	5	4	5	2	3	4	
9	Anjali Patil	38	House No 114, Civil Hospital Road, Sangli	2	1	1	1	4	4	1	1	1	2	
10	Tanmay Bhosale	41	House No 154, Madhavnagar Road, Sangli	4	1	2	1	3	3	2	1	5	5	
11	Ritika More	35	House No 106, Gaon Bhag, Sangli	2	1	1	3	3	5	2	2	5	2	
12	Yash Khot	47	House No 88, Miraj Road, Sangli	1	3	5	1	5	5	1	4	4	4	
13	Pooja Kadam	38	House No 196, Wanlesswadi, Sangli	2	5	2	4	3	2	1	3	5	2	
14	Harshal Patil	23	House No 188, Kupwad MIDC, Sangli	4	4	3	2	2	4	1	2	5	4	
15	Vaishnavi Mane	56	House No 153, Civil Hospital Road, Sangli	1	4	1	3	3	5	2	2	1	2	
16	Akshay Ghorpade	60	House No 175, Wanlesswadi, Sangli	3	1	5	3	5	2	3	2	5	3	
17	Megha Salunkhe	34	House No 173, Civil Hospital Road, Sangli	1	1	1	5	4	1	2	4	5	2	
18	Saurabh Kamble	30	House No 76, Madhavnagar Road, Sangli	2	3	5	5	1	3	1	5	4	4	
19	Komal Patil	48	House No 38, College Corner, Sangli	2	1	4	1	4	5	2	5	3	4	
20	Nikhil Desai	48	House No 151, Vishrambag, Sangli	2	1	3	1	1	3	2	5	4	4	
21	Sanjana Shetty	53	House No 172, Siddhivinayak Nagar, Sangli	4	5	5	5	4	3	1	4	1	4	
22	Pratik Sawant	33	House No 200, Wanlesswadi, Sangli	3	2	5	1	1	5	3	3	1	3	
23	Shruti Chougule	39	House No 73, 100 Feet Road, Sangli	2	2	1	4	5	2	5	1	4	2	
24	Kunal Patankar	45	House No 160, Sangli-Miraj Road, Sangli	4	1	5	5	5	3	1	3	4	3	
25	Isha Patil	18	House No 40, Kupwad MIDC, Sangli	4	2	5	4	4	1	3	4	4	1	
26	Ramesh Shinde	58	House No 123, Sangli-Miraj Road, Sangli	4	1	2	5	5	3	4	5	4	5	
27	Mahesh Patil	35	House No 192, 100 Feet Road, Sangli	1	5	5	3	5	3	5	3	5	2	
28	Sunita Jagtap	57	House No 30, Vishrambag, Sangli	1	2	5	2	5	3	2	5	2	4	
29	Vikas Jadhav	22	House No 47, Sangli-Miraj Road, Sangli	2	5	2	5	2	2	5	1	5	1	
30	Deepa Kulkarni	35	House No 179, Wanlesswadi, Sangli	3	2	1	1	4	1	4	1	4	3	
31	Rahul Patil	49	House No 171, Madhavnagar Road, Sangli	2	5	1	5	3	5	5	5	4	1	
32	Manisha Pawar	42	House No 166, Civil Hospital Road, Sangli	2	3	3	5	3	2	4	3	3	5	
33	Vijay Chavan	26	House No 134, Vishrambag, Sangli	2	1	1	4	2	4	5	2	1	3	
34	Swarnil Bhosale	29	House No 7, Kupwad MIDC, Sangli	5	1	2	1	2	4	3	4	3	4	
35	Ashwini Kadam	23	House No 20, Civil Hospital Road, Sangli	2	1	3	1	1	4	5	4	1	5	
36	Sandeep Mane	27	House No 174, Khanbhag, Sangli	4	4	5	5	4	4	5	1	2	4	
37	Gaurav Patil	29	House No 130, Miraj Road, Sangli	5	1	2	5	3	5	4	3	3	5	
38	Sheetal More	56	House No 21, Shivaji Nagar, Sangli	1	4	4	4	1	3	4	3	1	2	
39	Aniket Salunkhe	26	House No 198, Ramnagar, Sangli	3	2	4	4	1	1	5	3	5	2	

Quantitative Analysis:

The majority of respondents rely on multi-modal transportation. A significant percentage experiences delays and stress, confirming that fragmented transportation systems create inefficiency and unpredictability. The strong demand for an integrated platform highlights the need for a coordinated and predictive urban mobility solution.

Qualitative Analysis:

Common themes observed from open-ended responses:

- Missed connections increase travel time.
- Lack of synchronized real-time updates.
- Small delays disrupt the entire journey.
- Commuting causes mental stress and uncertainty.

Graphical Analysis:

The bar graph above represents:

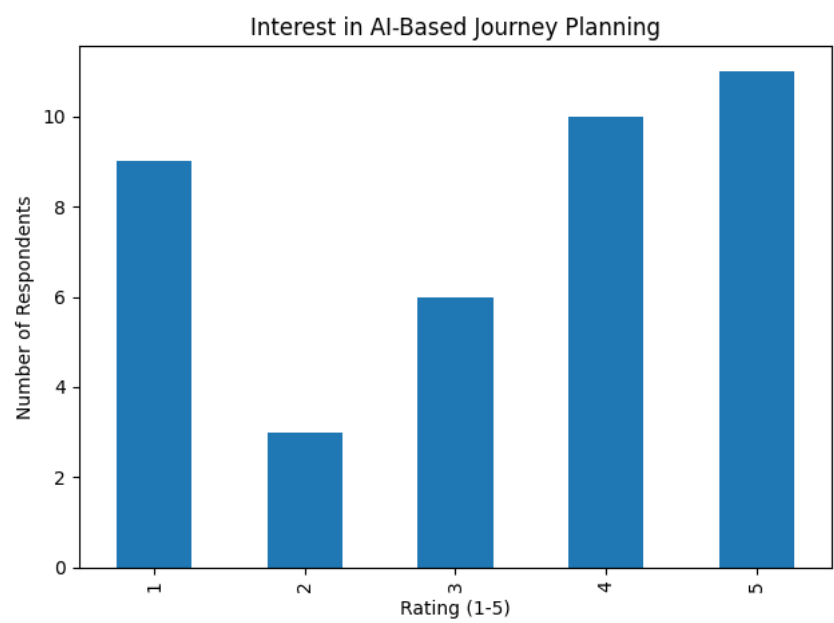
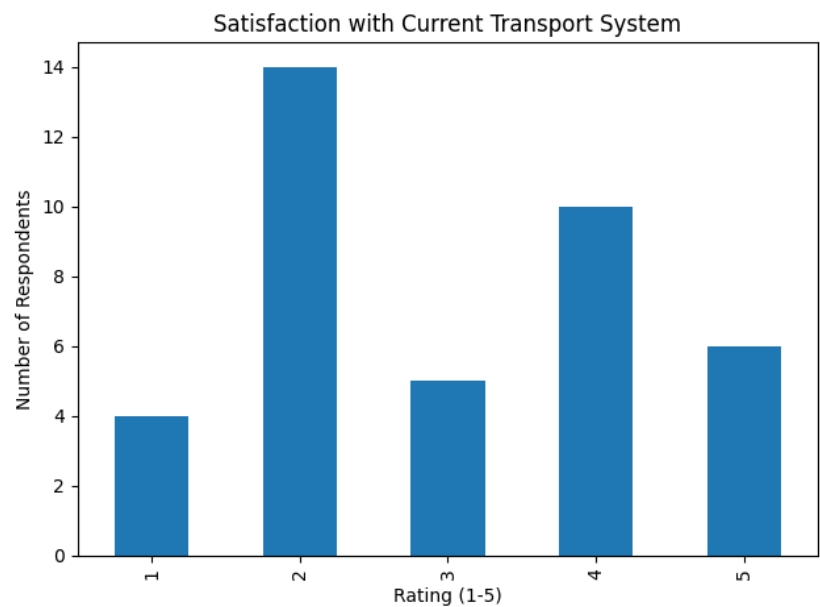
- Multi-modal transport usage
- Frequency of delays
- Stress levels among commuters
- Demand for integrated platform

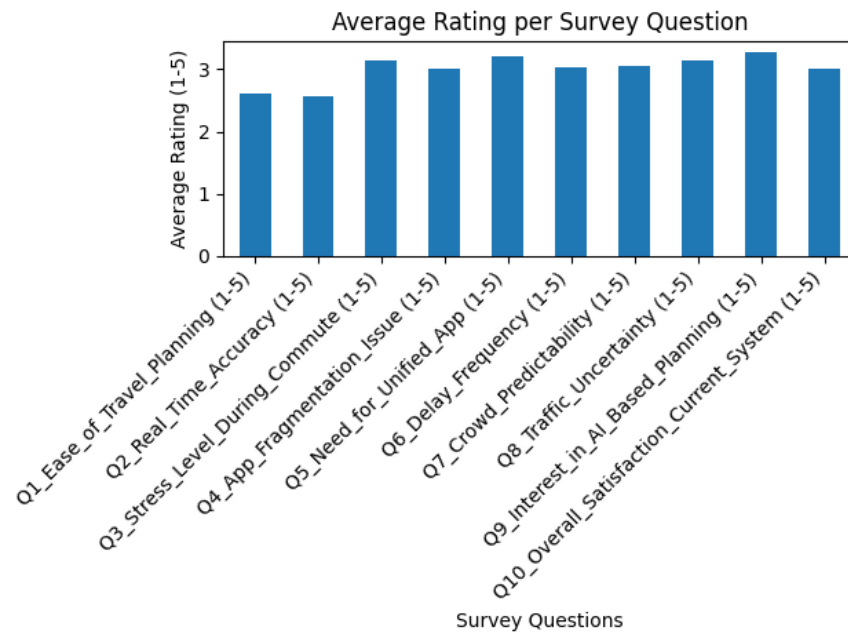
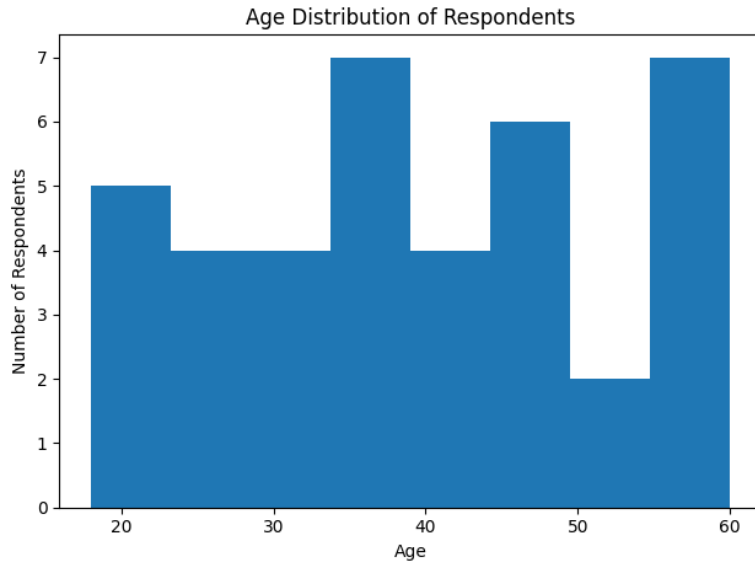
The graphical representation clearly shows high stress levels and strong demand for system integration.

The visuals show:

1. **Average Ratings per Question** – Clearly indicates high demand for a unified app and AI-based planning.
2. **Age Distribution** – Respondents are well distributed between 18–60.
3. **Interest in AI-Based Planning** – Strong positive inclination toward AI solutions.
4. **Satisfaction with Current System** – Mixed-to-low satisfaction, validating your problem statement.

Images of Our Analysis :





Submitted By:

Name :	Tanisha Borgave, Atharv Kulkarni, Ronakkumar Hurkat
Banch:	Computer Science & Engineering (Un-Aided)
Batch:	S5
PRN No:	245109072, 245109074, 245109078

Assignment 3

Title: Persona Development

Objective:

To construct detailed personas representing typical users to address real-world problems effectively.

Theory:

Persona development involves crafting detailed profiles for fictional characters that embody key traits of your target audience. Each persona should encapsulate the user's educational background, lifestyle, interests, values, objectives, requirements, desires, attitudes, and behaviors. These personas are not just static descriptions but are designed to realistically reflect the diversity within customer segments. Assigning a name and fleshing out a backstory enhances the realism and relatability of each persona.

Here are Some Personas for our Ps:

User Persona 1

Aarav Mehta

The Young Techie Commuter

Age: 24

Job: Software Engineer

Location: Pune

"I just want a
guaranteed,
stress-free
commute."



Goals

- On-time Office Arrival
- Save Travel Cost

Needs

- Real-time Alerts
- Multimodal Routing
- Auto Replanning

Scenario

Auto delayed → App alerts + alternate bus route

Needs

- 📶 Real-time Alerts
- 🚗 Multimodal Routing
- 🛡️ Auto Replanning

Frustrations

- 🕒 Missed Connections
- 🚗 Multiple Apps
- 📶 No ETA Accuracy

User Persona 2

Neha Sharma

The Corporate Professional

"I need comfort, predictability, and peace of mind." 12

Goals

- Stress-Free Travel
- Punctual Schedule
- Comfort & Safety

Scenario

- ☔ Rain Alert → 🚚
- 📱 App plans cab + train →
- 📅 Smooth & comfy journey

Needs

- ETA Confidence Score
- Crowd Prediction
- ☁️ Weather Backup Plans

Needs

- 👤 Unpredictable Crowds
- 🚗 Last-mile Stress

Frustrations

- 👤 Unpredictable Crowds
- 🚗 Daily Chaos

User Persona 3

Ramesh Patil

The Local Business Owner

Age: 40

Job: Business Owner

Location: Sangli

“Show me the fastest way, keep it simple!”

Goals

- Save Time & Fuel
- Avoid Traffic Delays
- On-time Deliveries

Scenario

 Traffic Alerts

 Simple UI (Local Lang)

 Voice Notifications

Needs

... Traffic Alerts

 Simple UI (Local Lang)

 Voice Notifications

Needs

 Roadblock Alert → Alternate Route →



User Persona 4

Vikram Desai

The Frequent Flyer / Executive

“Every minute matters –
I need **total reliability**.”

Age: 45

Job: Management
Consultant

Education: 

MBA (International
Business)

Location: Bengaluru

Scenario

- ✓ Zero Missed Flights
- ✚ Seamless Transfers
- ⚙️ Max Productivity

Needs

- 📅 High-Confidence ETA
- 👛 Premium Options
- 🔍 Auto Fallbacks

Needs

- 📱 Traffic Alerts
- 🗣️ Simple UI (Local Lang)
- 🔊 Voice Notifications

Frustrations

- 🚧 Traffic Risk
- 🕒 Missed Meetings
- 👥 Bad Coordination

Conclusion

The survey-based persona analysis clearly highlights the diverse commuting challenges faced by different urban user groups within the multimodal transportation ecosystem. Young tech professionals primarily seek time efficiency, real-time synchronization, and cost optimization. Corporate commuters demand predictability, comfort, and risk-aware journey planning. Local business owners prioritize simplicity, reliable traffic alerts, and minimal operational delays. Frequent travelers and executives require high-confidence travel coordination with automatic fallback mechanisms to avoid critical disruptions.

The statistical insights gathered from the survey demonstrate that a significant percentage of commuters experience missed connections, unpredictable delays, fragmented app usage, and high levels of travel-related stress. Most respondents expressed dissatisfaction with managing multiple transport apps independently and emphasized the need for unified, predictive journey planning rather than static route suggestions.

These findings validate the identified problem statement and strongly support the necessity for an intelligent, end-to-end mobility orchestration platform.

Therefore, **FlowTrip – Unified Urban Mobility Intelligence** effectively addresses these gaps by integrating multimodal transport systems into a single coordinated platform. Through AI-based delay prediction, risk modeling, real-time synchronization, confidence scoring, and automatic rerouting, FlowTrip transforms commuting from a fragmented and stressful process into a predictable, optimized, and user-centric experience.

The platform not only connects transport modes but also enhances reliability, reduces uncertainty, and builds commuter confidence within the urban mobility ecosystem.

Submitted By:

Name :	Tanisha Borgave, Atharv Kulkarni, Ronakkumar Hurkat
Branch:	Computer Science & Engineering (Un-Aided)
Batch:	S5
PRN No:	245109072, 245109074, 245109078

Experiment No 4

Title: Designing an Empathy Map Objective:

To create empathy maps for different commuter personas associated with the problem statement in order to better understand their behaviors, thoughts, emotions, and challenges within the urban mobility ecosystem.

Theory: An empathy map is a design thinking tool used to visualize and analyze a user's experience. It helps in understanding the user's perspective by categorizing insights into four main quadrants:

- **Says** - What the user explicitly expresses.
- **Thinks** - What the user thinks internally but may not say aloud.
- **Does** - The actions and behaviors performed by the user.
- **Feels** - Emotional responses during interaction.

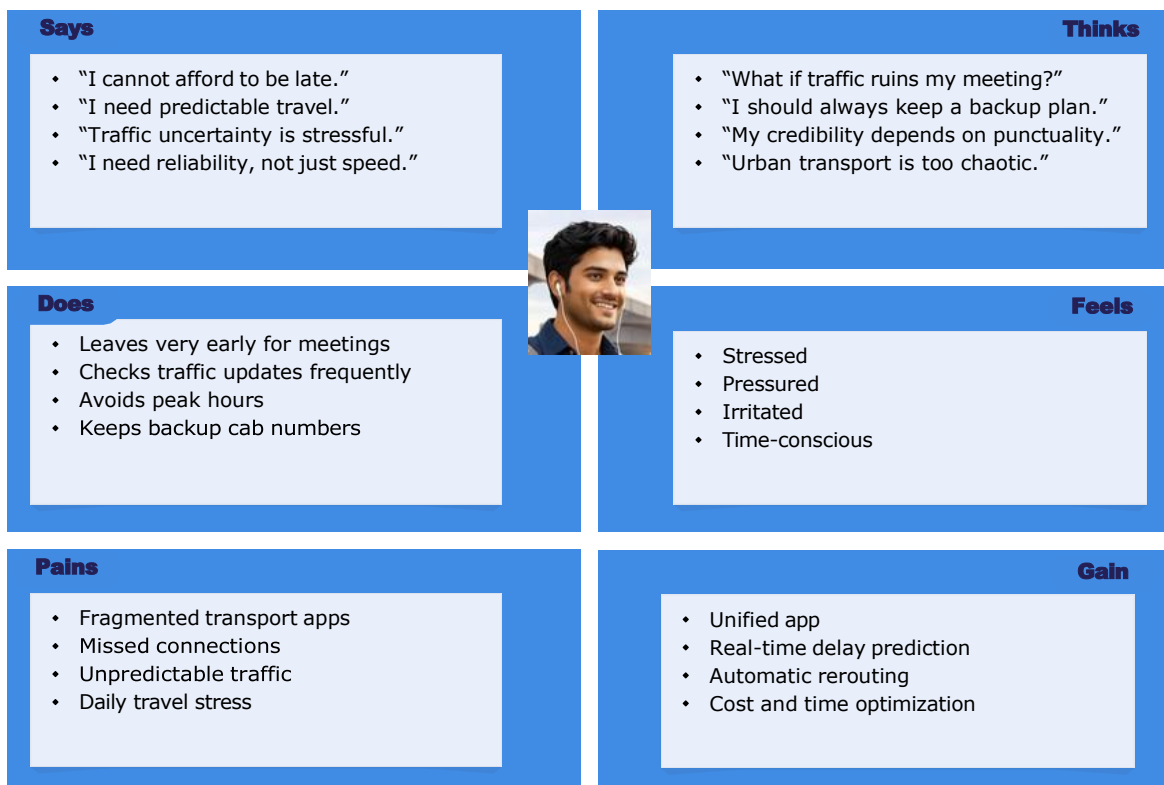
In addition to these quadrants, empathy maps often include:

- **Pain** - Frustrations, fears, and obstacles.
- **Gain** - Needs, expectations, and measures of success.

Empathy mapping helps identify contradictions between what users say and what they actually feel or do. This structured understanding supports better product design and user-centered decision-making.

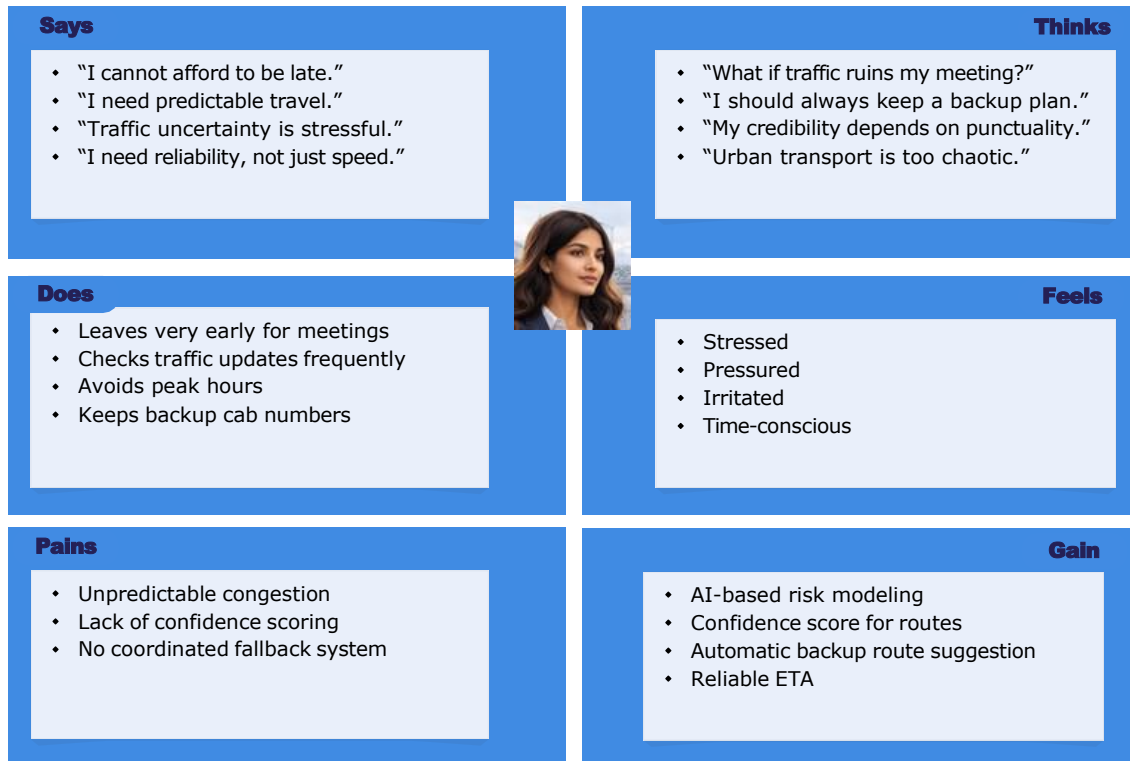
Empathy Map - Persona 1

Persona : Aarav Mehta (Young Tech Professional)



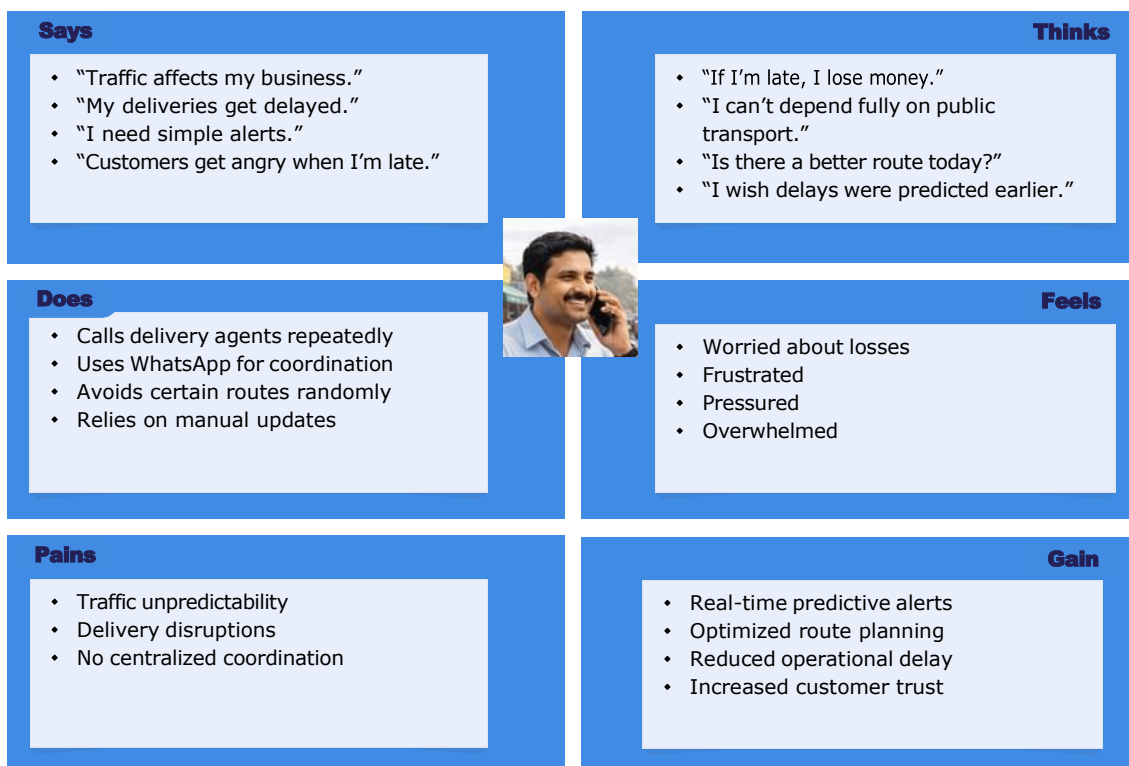
Empathy Map - Persona 2

Persona : Neha Sharma (The Corporate Professional)



Empathy Map - Persona 3

Persona : Ramesh Patil (Local Business Owner)



Empathy Map - Persona 4

Persona : Vikram Desai (The Frequent Flyer)



Conclusion: The empathy maps highlight emotional and behavioral gaps in the urban mobility ecosystem. Young professionals experience anxiety and inefficiency due to fragmented applications. Corporate commuters prioritize reliability and risk-aware planning to maintain credibility. Local business owners struggle with operational delays and financial uncertainty.

These insights validate the need for a unified, AI-powered mobility orchestration platform like FlowTrip that integrates multimodal transport, provides predictive intelligence, reduces uncertainty, and enhances commuter confidence.

Submitted By:

Name :	Tanisha Borgave, Atharv Kulkarni, Ronakkumar Hurkat
Branch:	Computer Science & Engineering (Un-Aided)
Batch:	S5
PRN No:	245109072, 245109074, 245109078

Experiment 5

Title: Customer Journey Map Development

Objective:

To develop a customer journey map that traces the interactions between a customer and an organization over a defined period, utilizing multiple channels to pinpoint improvements in user experience design.

Theory:

Customer journey maps visually depict the relationship between customers and an organization across various channels and over time. These maps are crucial for design teams to understand how customer experiences align with or deviate from their expectations, providing a structured way to identify necessary design enhancements.

Detailed Overview:

Customer journey maps are sophisticated tools based on thorough research. They encapsulate the full spectrum of customer interactions, mapping out the progression of experiences customers undergo. By employing these maps, companies gain valuable insights into their target audience's behaviors and preferences. The typical elements of a customer journey map include:

1. **Timeline:** Define the duration of the customer's journey, such as the first week of product use, capturing phases from initial awareness to potential repeat engagement or loyalty.
2. **Scenarios:** Outline the specific sequences of events the customer undergoes to achieve their goals, such as purchasing a ticket through a mobile app—from recognizing the need to the confirmation of purchase.
3. **Touchpoints:** Identify critical points of interaction between the customer and the business, noting what actions are taken and how they are executed.
4. **Channels:** Document the platforms (e.g., website, social media, in-store) where interactions occur, providing a clear view of where engagements take place.
5. **Emotions and Thoughts:** Integrate insights into what customers are thinking and feeling at each stage of their journey, which is crucial for empathizing with their experience.

Integration of How/Why Ladders:

How/Why Ladders can significantly enhance the development of customer journey maps by delving deeper into the reasons behind customer behaviors at each touchpoint. This technique involves:

- **Exploring Motivations:** Use How/Why Ladders to ask sequential questions that uncover deeper insights into customer motivations and barriers at each stage of the journey.
- **Refining Touchpoints and Scenarios:** By understanding the 'how' and 'why' behind customer actions, teams can more accurately define scenarios and touchpoints that reflect real customer challenges and needs.

Steps to Create a Customer Journey Map:

1. **Establish Objectives:** Clearly define the business goals and user experiences the map will address.
2. **Gather Data:** Conduct comprehensive research using interviews, surveys, and data analytics to collect experiences at all customer touchpoints.
3. **Identify Touchpoints and Channels:** List and review all customer interactions and channels, looking to expand or refine these based on insights gathered.
4. **Utilize an Empathy Map:** Construct an empathy map to visualize what the customer does, thinks, feels, and hears within their journey. This helps in understanding their needs and emotional responses.
5. **Draft the Map:** Compile the data into a visual map that outlines the customer's path through various touchpoints over the established timeline, including emotional states.
6. **Iterate and Enhance:** Continuously refine the map based on feedback and new insights to accurately represent the customer journey.
7. **Engage Stakeholders:** Share the finalized map with stakeholders across the organization to align on customer-focused strategies and improvements.

Customer Journey Maps (CJMs) are visual representations of the process a customer goes through to achieve a goal with your product or service. They are vital tools in understanding the user experience, identifying key interactions, and pinpointing areas that need improvement. The use of How/Why Ladders in creating these maps adds a layer of depth to the understanding of user motivations and decision-making processes.

- **How/Why Ladders and Customer Journey Maps**

How/Why Ladders are a methodological tool used to delve deeper into the user's cognitive and emotional processes by asking a series of escalating "how" and "why" questions. This technique helps to uncover not just what users are doing but also why they are doing it in terms of their deeper needs and motivations.

Application in Customer Journey Mapping

1. **Identifying Touchpoints:**
 - **How/Why Ladders** help to detail each step a user takes from the initial point of contact with a product or service to the final interaction. By thoroughly questioning each step, teams can identify all relevant touchpoints—where users interact with the product/service. This detailed interrogation helps to ensure no critical touchpoints are overlooked.

2. Understanding User Motivations:

- Each step and decision within a customer journey is driven by underlying motivations. The sequential questioning of How/Why Ladders allows teams to uncover these motivations, providing a clearer picture of what drives users to move from one stage of their journey to the next. This understanding can reveal the user's priorities, concerns, and expectations.

3. Highlighting Pain Points and Pleasures:

- By analyzing the answers to "how" and "why" questions, teams can identify not only the pain points (where users feel frustration or dissatisfaction) but also the pleasure points (where users feel satisfied or delighted). These insights are crucial for designing interventions that alleviate pain points and enhance pleasure points.

4. Enhancing Persona Integration:

- The insights gathered through How/Why Ladders can enrich user personas, making them more dynamic. Integrating these personas into the customer journey map allows teams to view the journey from the perspective of different user types, tailoring the experience to diverse user needs and behaviors.

5. Refining the Journey Phases:

- Customer journeys often encompass various phases such as awareness, consideration, purchase, and loyalty. How/Why Ladders help in clearly defining what happens in each of these phases and why users transition from one phase to another, allowing for more precise interventions designed to move users smoothly along their journey.

Benefits of Using How/Why Ladders in CJMs

- **Deep Insights:** Provides a deep, psychologically and emotionally informed understanding of the user experience.
- **User-Centric Design:** Supports the creation of more user-centric designs by focusing on real user needs and behaviors.
- **Strategic Development:** Helps in making strategic decisions about product development and marketing based on user insights.
- **Enhanced Communication:** Improves communication within the team and with stakeholders by clearly showing why certain design decisions are made based on user data.

Follow-up Actions:

Once the journey map is implemented, regularly monitor its impact through key performance indicators (KPIs) to measure improvements in customer experience and satisfaction. Adjust and update the map as needed to keep it relevant and reflective of actual customer experiences.

By integrating How/Why Ladders with the creation of customer journey maps, organizations can achieve a more profound understanding of customer needs, leading to more effective and empathetic user experience strategies.

Customer Journey Map

Persona: Aarav Mehta

Timeline: From planning commute at home to reaching office

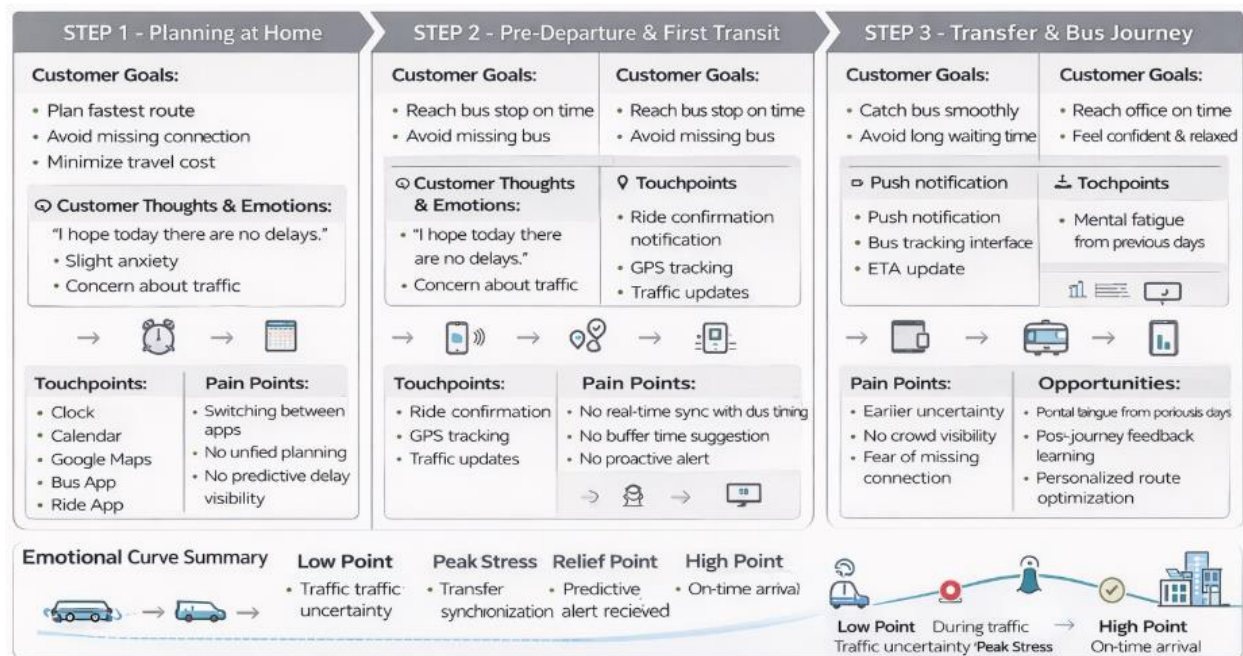
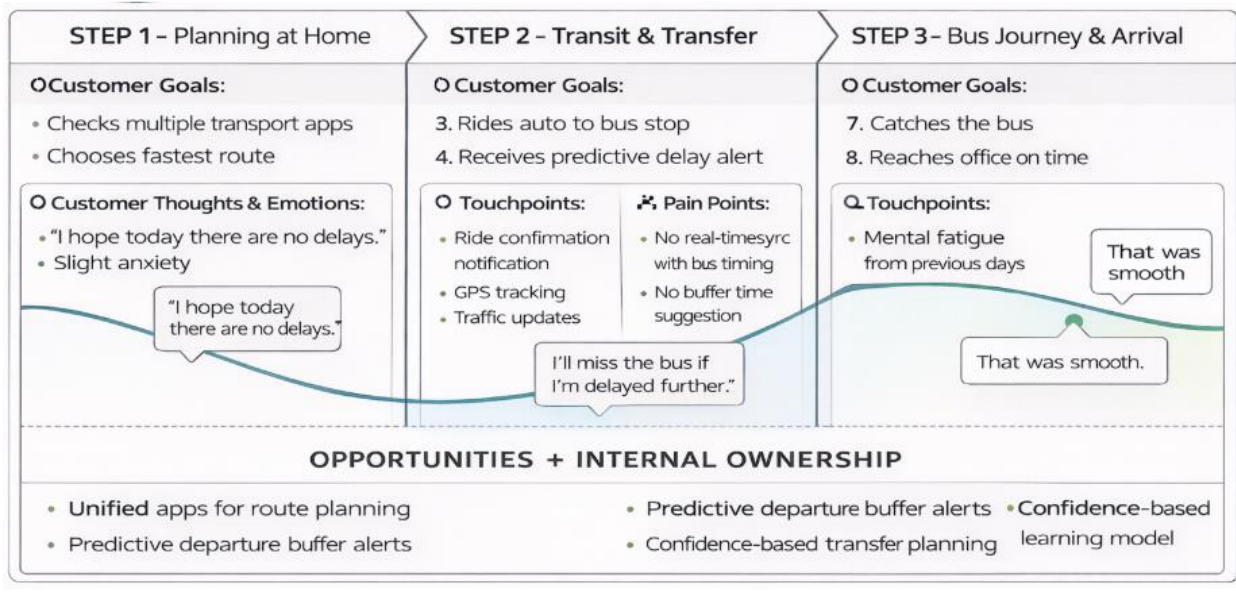
Emotional Curve Insight

- **Emotional Low:** During auto delay uncertainty
- **Emotional High:** When predictive alert prevents missed connection
- **Stress Peak:** Transfer point
- **Relief Point:** Synchronized connection confirmation

Journey Map in Table Format :

Phase	User Actions	Touchpoints	Thoughts	Emotions	Pain Points	Opportunities (FlowTrip Solution)
1. Awareness	Realizes he must leave for office	Clock, Calendar, Previous commute memory	"I hope today there are no delays."	Slight anxiety	Past experience of missed connections	• Show predictive daily travel risk summary
2. Planning	Opens multiple apps to check	Google Maps, Bus App, Ride App	"Which route is safer today?"	Confused	Switching between apps	• Unified multimodal planning in one app
3. Route Selection	Selects shared auto + bus route	Mobile app interface	"Will this auto reach on time?"	Uncertain	No buffer time guidance	• AI-based buffer prediction & confidence score
4. Pre-Departure	Waits for auto confirmation	Ride notification	"Traffic seems heavy..."	Worried	No predictive traffic alert	• Real-time traffic forecasting + leave-early suggestion
5. Transit – First Leg	Travels in auto	GPS tracking	"If I get delayed by 5 mins, Increasing stress I'll miss bus."		No sync with bus timing	• Transfer synchronization engine
6. Mid-Transit Alert	Receives predictive delay alert	Push notification	"App says bus ETA updated"	Normally no proactive alert	Normally no proactive alert	• Automatic alternate bus suggestion
7. Transfer	Reaches bus stop	Bus tracking interface	"Good, I made it."	Calm	Earlier uncertainty	• Confidence-based transfer planning
8. Arrival	Reaches office on	Final journey summary	"That was smooth."	Satisfied	Earlier mental fatigue	• Post-journey feedback learning model

Journey Map in Image:



How/Why Ladder Analysis

Why does Aarav feel daily stress - Because transfers are unpredictable.

Why are transfers unpredictable? - Transport systems are not synchronized.

Why are they not synchronized? - Each app works independently.

Why is that a problem? - Minor delay causes chain reaction.

Root Insight: Commuters need coordinated journey intelligence, not isolated route suggestions.

Key Design Opportunities Identified

1. Predictive departure timing
 2. Confidence score per segment
 3. Dynamic rerouting during disruptions
 4. Transfer synchronization
 5. Real-time crowd and delay modeling
-

Conclusion from Journey Map

The Customer Journey Map clearly shows that the highest emotional stress occurs during transfer synchronization and delay uncertainty. By introducing predictive risk modeling and real-time multimodal coordination, FlowTrip transforms commuting from a fragmented experience into a seamless, confidence-driven journey.

Submitted By:

Name :	Tanisha Borgave, Atharv Kulkarni, Ronakkumar Hurkat
Branch:	Computer Science & Engineering (Un-Aided)
Batch:	S5
PRN No:	245109072, 245109074, 245109078

Experiment No 6

Title: Defining the Problem and Ideation

Objective:

To define a meaningful, human-centered, and actionable problem statement based on user observations and to generate innovative ideas using Design Thinking methods.

Theory

An integral part of the Design Thinking process is defining a meaningful and actionable problem statement. This stage requires synthesizing observations gathered during the Empathize phase. A well-defined problem statement provides clarity, focus, and direction for the Ideation stage. If the problem is poorly defined, the design process lacks structure and may result in ineffective solutions.

In this experiment, the problem is analyzed from a human-centered perspective, ensuring it is broad enough to allow creative freedom while being narrow enough to remain manageable.

Observations from the Empathize Stage

Based on user behavior and commuting experiences:

What Users Said

- “I have to check multiple apps to plan one journey.”
- “If one bus is late, my whole schedule gets disturbed.”
- “I never know if I will reach on time.”

What Users Did

- Constantly switch between transport apps.
- Frequently check for updates and delays.
- Leave earlier than necessary to avoid uncertainty.

What Users Thought

- “There must be a simpler way to manage this.”
- “I hope nothing gets delayed today.”
- “I can’t rely completely on public transport timing.”

What Users Felt

- Stressed and anxious about missing connections.
- Mentally exhausted from daily planning.
- Frustrated due to unpredictable disruptions.

Space Saturate and Group (Synthesis)

After clustering insights from observations, the following themes emerged:

- Fragmented information systems
- Lack of synchronization between transport modes
- High dependency on manual planning
- Reactive rather than predictive updates
- Increased mental effort and travel anxiety

These grouped insights highlight a clear gap in integrated and intelligent urban mobility coordination.

Problem Statement (Human-Centered and Action-Oriented)

Create an integrated and intelligent urban mobility system that enables commuters to experience seamless, end-to-end journey coordination because current transportation systems operate in isolated segments, forcing commuters to manually manage schedules, transfers, and disruptions, which leads to stress, uncertainty, and inefficiency.

Point of View (POV)

Urban daily commuters need a unified and predictive journey management system because disconnected transport services and fragmented information make commuting unpredictable and mentally exhausting.

Why–How Laddering

Why is commuting stressful?

Because commuters must manually coordinate multiple transport modes.

Why must they coordinate manually?

Because transport systems operate independently without unified synchronization.

Why is there no synchronization?

Because existing systems focus on individual segments rather than the entire journey.

How can this be solved?

By designing an integrated, predictive, and real-time coordinated mobility platform that connects all travel segments into one seamless experience.

How Might We (HMW) Questions

1. How might we integrate different modes of urban transport into one coordinated system?
2. How might we provide predictive alerts before delays affect commuters?
3. How might we reduce the mental load of planning daily travel?
4. How might we ensure end-to-end journey visibility across all travel segments?
5. How might we automatically suggest optimized alternative routes during disruptions?

Conclusion

Through the Define stage of the Design Thinking process, we synthesized user observations into a clear, human-centered, and actionable problem statement. The identified challenge focuses on transforming fragmented urban commuting into a seamless, predictable, and stress-free experience. The formulated POV and How Might We questions provide a strong foundation for the Ideation stage, where innovative and practical solutions can be developed.

Submitted By:

Name :	Tanisha Borgave, Atharv Kulkarni, Ronakkumar Hurkat
Branch:	Computer Science & Engineering (Un-Aided)
Batch:	S5
PRN No:	245109072, 245109074, 245109078

Experiment no 7

Title: Ideation

Objective: To generate ideas to solve the problem.

Theory:

Ideation is the creative stage in the Design Thinking process where participants engage in generating a multitude of ideas to address a well-defined problem. Positioned as the third phase, following “Empathize” (where insights are gathered from user research and observation) and “Define” (where these insights are synthesized into a coherent problem statement), ideation is crucial for developing innovative solutions.

Understanding Ideation

Ideation sessions are essential for discovering new perspectives and innovative solutions. To facilitate effective ideation, it's vital to establish a conducive environment that encourages free thinking and open collaboration. This begins with a clear problem statement or a pointed question, such as “How might we design an app to locate budget-friendly hotels in safe areas?” This sets a clear direction and fosters productive ideation.

Setting Up for Success

Creating the right setting involves preparing a space that stimulates creativity. This might include displaying personas, pertinent data, and other relevant materials that keep the problem in focus. Establishing ground rules like a time limit, prioritizing the quantity of ideas over quality, and maintaining a no-distractions policy helps maintain focus and energy. Emphasizing that there are no bad ideas during these sessions encourages participation from everyone, fostering a culture where ideas can build upon one another.

Ideation Techniques

A variety of techniques can enhance ideation sessions:

- **Brain storming:** Participants build on each other’s ideas, enhancing and expanding them.
- **Brain dumping:** Similar to brainstorming but performed individually.
- **Brain writing:** Ideas are written down and passed around for others to add to before group discussion.
- **Brain walking:** An active form of brainwriting where participants move around the room, adding to others’ ideas.
- **Worst Possible Idea:** Encourages generating deliberately bad ideas to ease participation and can unexpectedly lead to useful concepts.

- **Challenging Assumptions:** Questions established beliefs to open up new avenues of thought.
- **Mind mapping:** A visual method to organize thoughts and ideas connected to the central problem.
- **Sketch storming:** Uses quick sketches to visualize ideas and explore potential solutions.
- **Story boarding:** Creates a visual narrative to explore the problem and potential solutions in a dynamic format.
- **SCAMPER:** Encourages looking at the problem through different actions like Substitute, Combine, Adapt, etc.
- **Analogies:** Drawing comparisons to better articulate ideas and solutions.
- **Provocation:** An extreme method of lateral thinking to break conventional thinking and explore novel territories.
- **Movement:** Taking a dynamic approach to physically overcome barriers and find new themes or solutions.
- **Cheat storming:** Utilizing previously developed ideas as stimuli for further creativity.
- **Crowd storming:** Engaging a broader audience through social media to generate and refine ideas.
- **Creative Pause:** Taking deliberate breaks to overcome mental blocks and refresh perspective.

Leveraging Ideation

Ideation is not just about creating ideas but about exploring the widest possible range of solutions. It's a process that combines the conscious and unconscious mind, blending logical and creative thinking. The process benefits immensely from structured yet flexible approaches that allow participants to navigate through creative challenges effectively. If obstacles arise, strategies such as "breaking the law" (questioning constraints), "stealing" (adapting ideas from other industries), reversing problems, and employing laddering techniques (shifting between abstract and concrete ideas) can provide alternative pathways.

By embracing these diverse methods, teams can ensure their ideation sessions produce a rich array of viable solutions, driving innovation forward.

Title of your case study: FlowTrip – Unified Urban Mobility Intelligence

Problem Statement:

Urban commuters face fragmented transport systems, unpredictable delays, missed connections, and high travel stress due to disconnected mobility applications and lack of predictive coordination. There is a need for an intelligent, unified, and risk-aware mobility platform that ensures synchronized, reliable, and optimized urban travel.

IDEATION:

To solve the identified problem, multiple ideation techniques were applied including:

- Brainstorming
- Mind Mapping
- SCAMPER
- Worst Possible Idea
- Challenging Assumptions
- Storyboarding

The ideation session focused on generating a wide range of innovative, technology-driven solutions to reduce commuter stress and improve travel reliability.

Ground Rules Followed:

- Quantity over quality
- No idea was rejected initially
- Ideas were built upon collaboratively
- Focus on solving emotional and functional user pain points

GENERATED IDEA'S:

Unified Multimodal Dashboard

A single platform integrating metro, bus, cab, auto, and flights with synchronized real-time updates.

AI-Based Delay Prediction Engine

Machine learning models that predict traffic congestion and transport delays before they happen.

Confidence Score for Every Route

Each suggested route displays a reliability score based on historical and real-time data.

Automatic Smart Rerouting

If delay risk increases, the system automatically suggests alternate optimized routes.

Airport Synchronization Module

For frequent flyers, integrates flight schedules with live city traffic and check-in buffer timing.

Risk Alert Notifications

Push notifications for high-delay probability, weather disruptions, or route instability.

Cost-Time Optimization Slider

Users can adjust preferences between:

- Fastest
- Cheapest
- Most reliable
- Balanced

Business Mode

Special mode for corporate commuters and business owners with:

- Meeting buffer calculation
- Delivery time prediction
- High-priority alerts

Stress Index Monitor

Measures user commute patterns and suggests safer or less stressful travel alternatives.

Backup Travel Recommendation

Automatic cab booking or alternate metro suggestions if primary route risk increases.

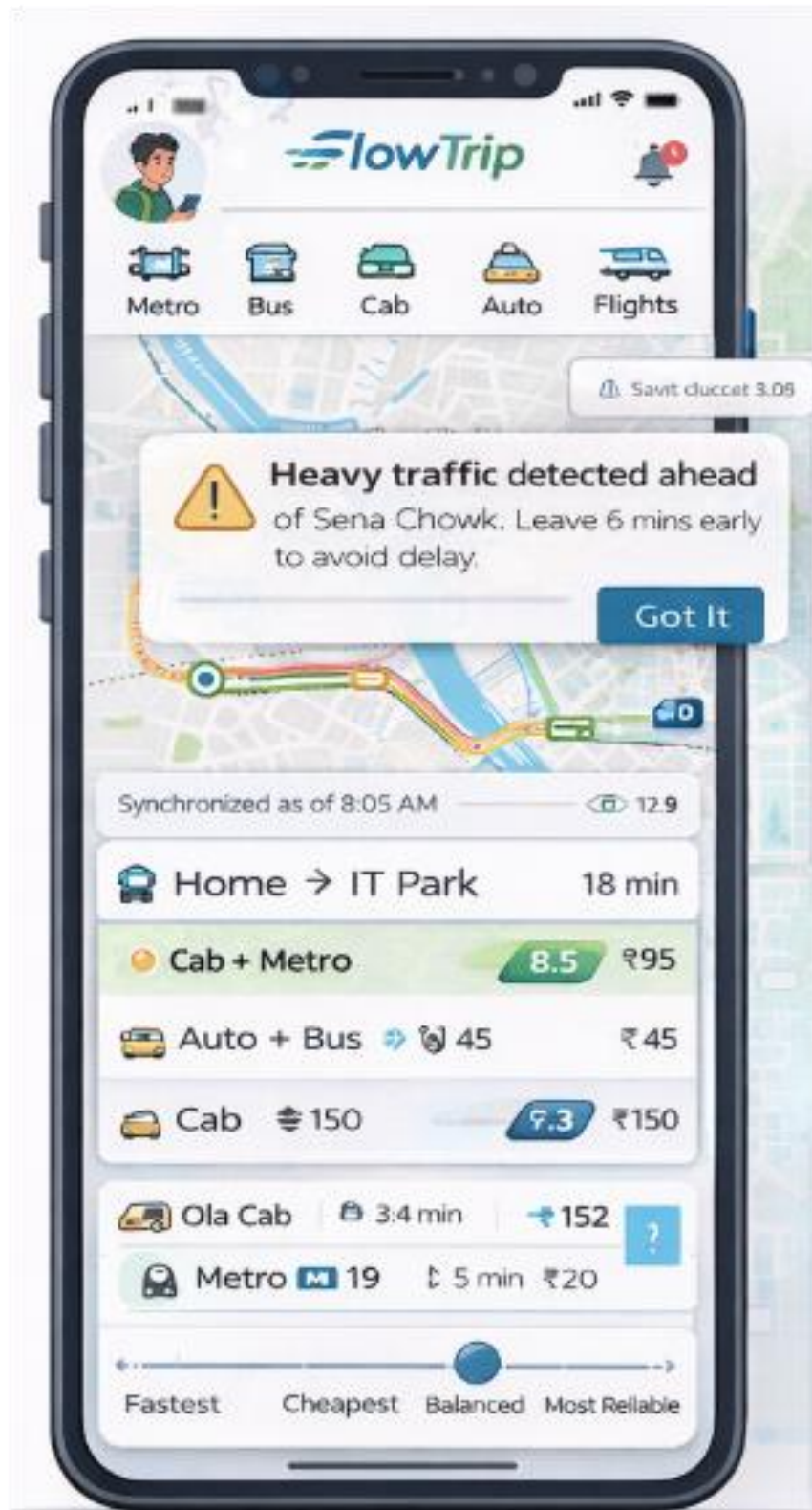
Community Traffic Intelligence

Users can report live disruptions to improve system learning.

Gamification for Smart Commuting

Rewards for using optimized and eco-friendly routes.

Here are some images of Ideas we have generated :





Submitted By:

Name :	Tanisha Borgave, Atharv Kulkarni, Ronakkumar Hurkat
Branch:	Computer Science & Engineering (Un-Aided)
Batch:	S5
PRN No:	245109072, 245109074, 245109078